2.26-3.4 Weekly Record

Continue removing non-food-related and too general `categories` , get `pa_cleaned_categories.json` (count: 195)

Only retain businesses that hold at least one matching category in `pa_cleaned_categories.json` (7209 businesses→ 6926 businesses)

Finding: Non-Dining Businesses in Present Data

supermarket, convenience store, etc. instead of restaurant, cafe, bar


```
{
  "business_id": "BIJz39GnlrVUhrWPiWGECQ",
  "name": "ALDI",
  "address": "1205 Macdade Blvd",
  "city": "Collingdale",
  "state": "PA",
  "postal_code": "19023",
  "latitude": 39.9091231251,
  "longitude": -75.2841462163,
  "stars": 3.5,
  "review_count": 14,
  "is_open": 1,
  "attributes": {
    "BusinessParking": "{'garage': False, 'street': False, 'validated': False, 'lot': True, 'valet': False}",
    "BusinessAcceptsCreditCards": "True",
    "RestaurantsDelivery": "True",
    "RestaurantsTakeOut": "True",
    "RestaurantsPriceRange2": "1",
    "BikeParking": "False"
  },
  "categories": "Beer, Wine & Spirits, Organic Stores, Grocery, Fruits & Veggies, Specialty Food, Food, Meat Shops, Depart
```


To filter out:

train a classifier ×

leverage a LLM/PLM, such as the locally deployed DeepSeek or T5

//Try the prompt once for one entry instead the whole set if using LLM

keyword-based filtering √

```
# Remove businesses if EXCLUDE_KEYWORDS are present and RESTAURANT_K
```

```
    if category_list & EXCLUDE_KEYWORDS and not category_list & RESTAURANT_
```

(6926 businesses→ 6616 businesses)

Categories Clustering

Word Embeddings: Group categories based on **semantic meaning**

Business Co-Occurrence: Find **real-world category relationships**

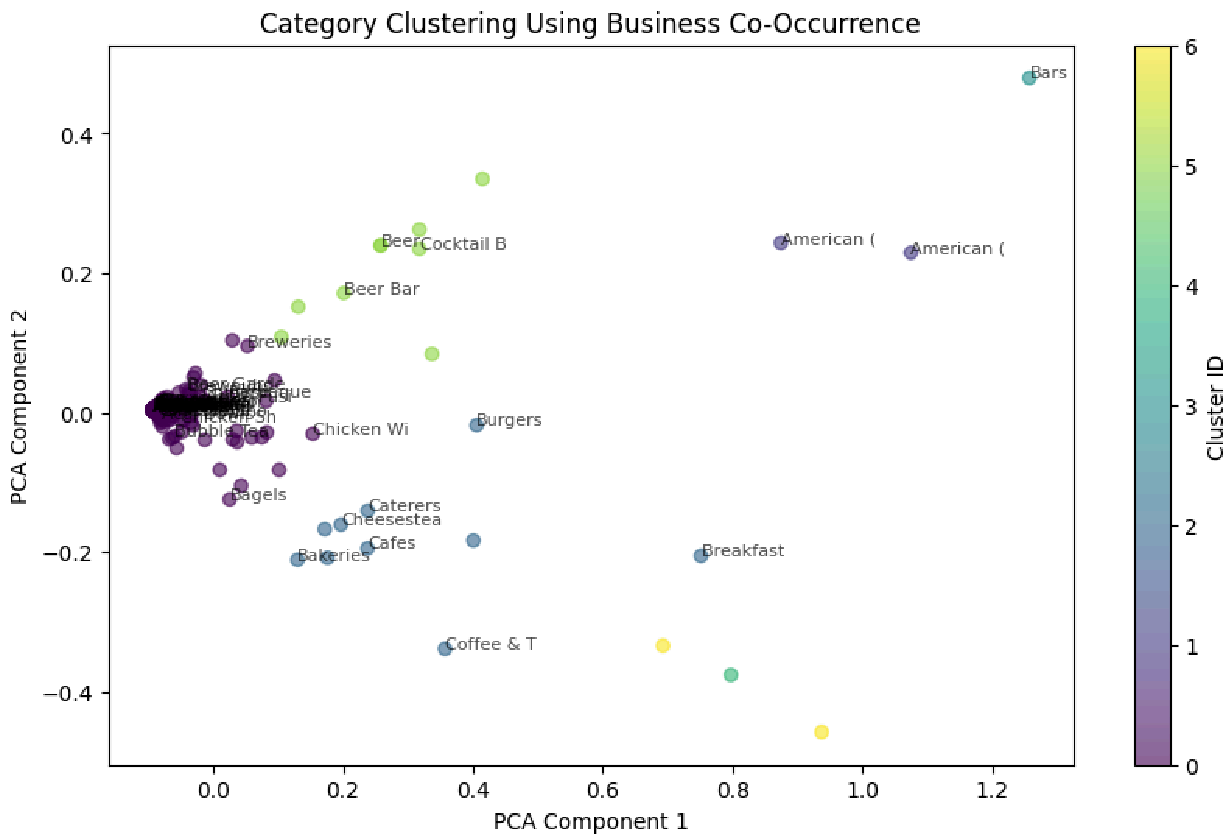
Cluster solely based on either is not accurate

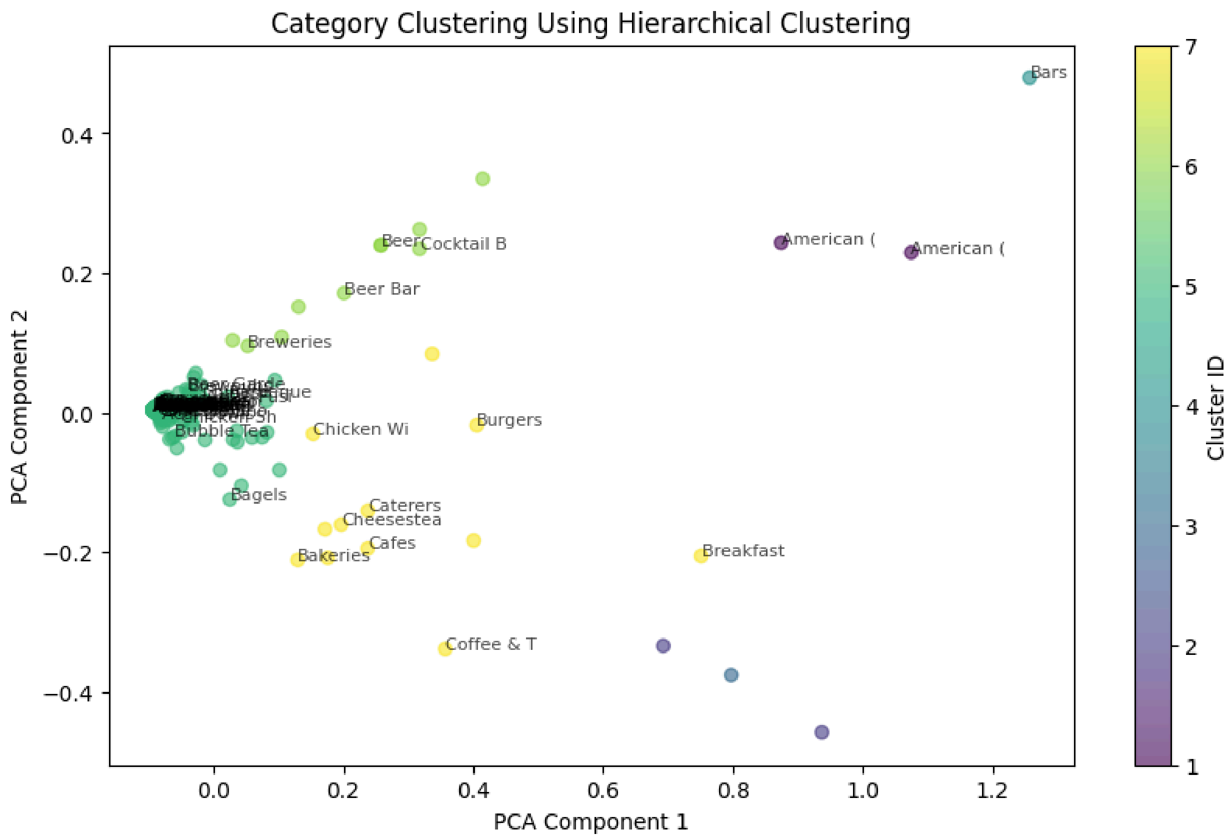
Work Flow: Cluster Categories Based on Business Co-Occurrence

1. Build a Category Co-Occurrence Matrix
2. Use elbow points/silhouette score/knee locator or *dendrogram* to determine best k (how many clusters)
3. Apply PCA and K-Means or *Hierarchical* Clustering on the Co-Occurrence Matrix

Imbalanced: some clusters seem **too broad** while others are **too small**

One Cluster is Too Large (128 Categories): It contains **almost everything**, including `"Vegan"`, `"Fast Food"`, `"Mexican"`, `"Sushi Bars"`, `"Thai"`, `"Tacos"`, `"Middle Eastern"`, etc.


Category Clustering Using Business Co-Occurrence
PCA Component 1
PCA Component 2
Cluster ID
Bars
American (
American (
Beer
Cocktail B
Beer Bar
Breweries
Burgers
Chicken Wi
Bubble Tea
Bagels
Caterers
Cheesestea
Cafes
Bakeries
Breakfast
Coffee & T



√ Hybrid Clustering Work Flow:

1. **Compute Co-Occurrence Clusters** → Capture how businesses **actually** use categories.
2. **Compute Word Embedding Clusters** → Capture **semantic relationships**.
3. **Merge Both Cluster Assignments**:
   - If two categories **co-occur often AND are semantically similar**, place them in the same cluster.

```
combined_vectors = np.hstack((semantic_vectors, co_occurrence_vectors))
```

Result: `category_clusters_hybrid.json` more structured and meaningful

- **Cluster 1** → **Latin American Cuisine** (Mexican, Brazilian, Cuban, Peruvian, etc.).
- **Cluster 2** → **Asian Cuisine** (Chinese, Japanese, Thai, Korean, etc.).
- Cluster 3 → Global & European Cuisine (Indian, Greek, Mediterranean, Russian, etc.).
- **Cluster 4** → **Western Classics** (American (New), American (Traditional), Italian, Pizza, Breakfast & Brunch).
- **Cluster 5 → all bars, wineries, and alcohol-related venues** (Beer Bar, Cocktail Bars, Pubs, Wine Bars, Breweries, Whiskey Bar, Dive Bars)
- Cluster 6 → Casual Street Food & Snacks (Noodles, Poke, Tacos, Sushi Bars, Salad, Falafel).
- Cluster 7 → Desserts, Cafés, and Fast Food (Bakeries, Bubble Tea, Desserts, Coffee & Tea, Fast Food, Steakhouses). **fast, take-away dining options?**

Process Attributes

Extracted 38 attributes based on 6616 businesses

Removed 11 low-occurrence attributes: threshold at 5000 (75% of 6616 businesses)

Fixed "BusinessParking" ("True" as long as there is one "True")

Next

How to integrate these clusters into recommendation system?

Further attributes processing (Wifi, Ambience, None Value)

//Check where `ambience` comes from: business itself or platform

Process `ambience`: pull out "true" ones and put them into "description"

// Check what the value of `RestaurantsPriceRange2` (1, 2 ,3 ,4) means

// Switch to writing (chapter of data processing)